

Coeff generate [-50,50]

Test 1, Time generate polynomial: 0.000513 s

Sturm suite [1]:

Degree 40:

Time: 0.000014 s

Coeff numerator bits: 14

Degree 39:

Time: 0.000017 s

Coeff numerator bits: 40

Sturm suite [2]:

Degree 39:

Time: 0.000017 s

Coeff numerator bits: 53

Degree 38:

Time: 0.000061 s

Coeff numerator bits: 104

Sturm suite [3]:

Degree 38:

Time: 0.000046 s

Coeff numerator bits: 150

Degree 37:

Time: 0.000117 s

Coeff numerator bits: 273

Sturm suite [4]:

Degree 37:

Time: 0.000106 s

Coeff numerator bits: 306

Degree 36:

Time: 0.000184 s

Coeff numerator bits: 505

Sturm suite [5]:

Degree 36:

Time: 0.000197 s

Coeff numerator bits: 578

Degree 35:

Time: 0.000266 s
Coeff numerator bits: 888

Sturm suite [6]:
Degree 35:
Time: 0.000285 s
Coeff numerator bits: 934

Degree 34:
Time: 0.000439 s
Coeff numerator bits: 1360

Sturm suite [7]:
Degree 34:
Time: 0.000434 s
Coeff numerator bits: 1455

Degree 33:
Time: 0.000560 s
Coeff numerator bits: 2018

Sturm suite [8]:
Degree 33:
Time: 0.000636 s
Coeff numerator bits: 2074

Degree 32:
Time: 0.000967 s
Coeff numerator bits: 2793

Sturm suite [9]:
Degree 32:
Time: 0.000995 s
Coeff numerator bits: 2900

Degree 31:
Time: 0.001194 s
Coeff numerator bits: 3787

Sturm suite [10]:
Degree 31:
Time: 0.001213 s
Coeff numerator bits: 3858

Degree 30:
Time: 0.001616 s
Coeff numerator bits: 4922

Sturm suite [11]:
Degree 30:
Time: 0.001740 s
Coeff numerator bits: 5039

Degree 29:
Time: 0.002990 s
Coeff numerator bits: 6297

Sturm suite [12]:
Degree 29:
Time: 0.002988 s
Coeff numerator bits: 6360

Degree 28:
Time: 0.002911 s
Coeff numerator bits: 7806

Sturm suite [13]:
Degree 28:
Time: 0.003043 s
Coeff numerator bits: 7929

Degree 27:
Time: 0.003938 s
Coeff numerator bits: 9574

Sturm suite [14]:
Degree 27:
Time: 0.004205 s
Coeff numerator bits: 9655

Degree 26:
Time: 0.005517 s
Coeff numerator bits: 11485

Sturm suite [15]:
Degree 26:
Time: 0.005101 s
Coeff numerator bits: 11610

Degree 25:
Time: 0.006479 s
Coeff numerator bits: 13653

Sturm suite [16]:
Degree 25:
Time: 0.006381 s
Coeff numerator bits: 13730

Degree 24:
Time: 0.007248 s
Coeff numerator bits: 15980

Sturm suite [17]:
Degree 24:
Time: 0.007600 s
Coeff numerator bits: 16092

Degree 23:
Time: 0.008988 s
Coeff numerator bits: 18529

Sturm suite [18]:
Degree 23:
Time: 0.009393 s
Coeff numerator bits: 18608

Degree 22:
Time: 0.011461 s
Coeff numerator bits: 21247

Sturm suite [19]:
Degree 22:
Time: 0.010886 s
Coeff numerator bits: 21358

Degree 21:
Time: 0.012994 s
Coeff numerator bits: 24185

Sturm suite [20]:
Degree 21:
Time: 0.013312 s

Coeff numerator bits: 24284

Degree 20:

Time: 0.014384 s

Coeff numerator bits: 27328

Sturm suite [21]:

Degree 20:

Time: 0.015521 s

Coeff numerator bits: 27438

Degree 19:

Time: 0.016880 s

Coeff numerator bits: 30683

Sturm suite [22]:

Degree 19:

Time: 0.016491 s

Coeff numerator bits: 30784

Degree 18:

Time: 0.019703 s

Coeff numerator bits: 34243

Sturm suite [23]:

Degree 18:

Time: 0.019197 s

Coeff numerator bits: 34336

Degree 17:

Time: 0.021258 s

Coeff numerator bits: 37992

Sturm suite [24]:

Degree 17:

Time: 0.021605 s

Coeff numerator bits: 38107

Degree 16:

Time: 0.025488 s

Coeff numerator bits: 41967

Sturm suite [25]:

Degree 16:

Time: 0.023227 s
Coeff numerator bits: 42039

Degree 15:
Time: 0.025396 s
Coeff numerator bits: 46074

Sturm suite [26]:
Degree 15:
Time: 0.025176 s
Coeff numerator bits: 46183

Degree 14:
Time: 0.027001 s
Coeff numerator bits: 50402

Sturm suite [27]:
Degree 14:
Time: 0.026690 s
Coeff numerator bits: 50451

Degree 13:
Time: 0.028320 s
Coeff numerator bits: 54828

Sturm suite [28]:
Degree 13:
Time: 0.028569 s
Coeff numerator bits: 54931

Degree 12:
Time: 0.031093 s
Coeff numerator bits: 59456

Sturm suite [29]:
Degree 12:
Time: 0.028338 s
Coeff numerator bits: 59504

Degree 11:
Time: 0.030830 s
Coeff numerator bits: 64187

Sturm suite [30]:

Degree 11:
Time: 0.031344 s
Coeff numerator bits: 64266

Degree 10:
Time: 0.031740 s
Coeff numerator bits: 69070

Sturm suite [31]:
Degree 10:
Time: 0.029074 s
Coeff numerator bits: 69105

Degree 9:
Time: 0.032475 s
Coeff numerator bits: 73990

Sturm suite [32]:
Degree 9:
Time: 0.029488 s
Coeff numerator bits: 74015

Degree 8:
Time: 0.031546 s
Coeff numerator bits: 78963

Sturm suite [33]:
Degree 8:
Time: 0.029455 s
Coeff numerator bits: 78937

Degree 7:
Time: 0.031441 s
Coeff numerator bits: 83873

Sturm suite [34]:
Degree 7:
Time: 0.029132 s
Coeff numerator bits: 83842

Degree 6:
Time: 0.030871 s
Coeff numerator bits: 88720

Sturm suite [35]:

Degree 6:

Time: 0.026263 s

Coeff numerator bits: 88660

Degree 5:

Time: 0.028034 s

Coeff numerator bits: 93442

Sturm suite [36]:

Degree 5:

Time: 0.024765 s

Coeff numerator bits: 93249

Degree 4:

Time: 0.027875 s

Coeff numerator bits: 97757

Sturm suite [37]:

Degree 4:

Time: 0.021513 s

Coeff numerator bits: 97558

Degree 3:

Time: 0.022185 s

Coeff numerator bits: 101676

Sturm suite [38]:

Degree 3:

Time: 0.017424 s

Coeff numerator bits: 101259

Degree 2:

Time: 0.018094 s

Coeff numerator bits: 104682

Sturm suite [39]:

Degree 2:

Time: 0.015692 s

Coeff numerator bits: 104081

Degree 1:

Time: 0.013694 s

Coeff generate [-100,100]

Time generate polynomial: 0.000561 s

Sturm suite [1]:

Degree 40:

Time: 0.000020 s

Coeff numerator bits: 14

Degree 39:

Time: 0.000022 s

Coeff numerator bits: 37

Sturm suite [2]:

Degree 39:

Time: 0.000024 s

Coeff numerator bits: 57

Degree 38:

Time: 0.000056 s

Coeff numerator bits: 107

Sturm suite [3]:

Degree 38:

Time: 0.000046 s

Coeff numerator bits: 159

Degree 37:

Time: 0.000101 s

Coeff numerator bits: 282

Sturm suite [4]:

Degree 37:

Time: 0.000129 s

Coeff numerator bits: 327

Degree 36:

Time: 0.000222 s

Coeff numerator bits: 545

Sturm suite [5]:

Degree 36:

Time: 0.000266 s

Coeff numerator bits: 620

Degree 35:

Time: 0.000349 s
Coeff numerator bits: 958

Sturm suite [6]:
Degree 35:
Time: 0.000364 s
Coeff numerator bits: 1018

Degree 34:
Time: 0.000494 s
Coeff numerator bits: 1499

Sturm suite [7]:
Degree 34:
Time: 0.000517 s
Coeff numerator bits: 1591

Degree 33:
Time: 0.000733 s
Coeff numerator bits: 2223

Sturm suite [8]:
Degree 33:
Time: 0.000763 s
Coeff numerator bits: 2301

Degree 32:
Time: 0.000936 s
Coeff numerator bits: 3113

Sturm suite [9]:
Degree 32:
Time: 0.000991 s
Coeff numerator bits: 3224

Degree 31:
Time: 0.001420 s
Coeff numerator bits: 4222

Sturm suite [10]:
Degree 31:
Time: 0.001516 s
Coeff numerator bits: 4315

Degree 30:
Time: 0.002130 s
Coeff numerator bits: 5524

Sturm suite [11]:
Degree 30:
Time: 0.002066 s
Coeff numerator bits: 5631

Degree 29:
Time: 0.002614 s
Coeff numerator bits: 7067

Sturm suite [12]:
Degree 29:
Time: 0.002834 s
Coeff numerator bits: 7167

Degree 28:
Time: 0.003402 s
Coeff numerator bits: 8804

Sturm suite [13]:
Degree 28:
Time: 0.003759 s
Coeff numerator bits: 8928

Degree 27:
Time: 0.004489 s
Coeff numerator bits: 10802

Sturm suite [14]:
Degree 27:
Time: 0.004801 s
Coeff numerator bits: 10903

Degree 26:
Time: 0.006760 s
Coeff numerator bits: 13004

Sturm suite [15]:
Degree 26:
Time: 0.006177 s
Coeff numerator bits: 13142

Degree 25:
Time: 0.007187 s
Coeff numerator bits: 15478

Sturm suite [16]:
Degree 25:
Time: 0.007321 s
Coeff numerator bits: 15588

Degree 24:
Time: 0.008795 s
Coeff numerator bits: 18181

Sturm suite [17]:
Degree 24:
Time: 0.008972 s
Coeff numerator bits: 18304

Degree 23:
Time: 0.010962 s
Coeff numerator bits: 21129

Sturm suite [18]:
Degree 23:
Time: 0.011413 s
Coeff numerator bits: 21240

Degree 22:
Time: 0.013707 s
Coeff numerator bits: 24293

Sturm suite [19]:
Degree 22:
Time: 0.013799 s
Coeff numerator bits: 24422

Degree 21:
Time: 0.016081 s
Coeff numerator bits: 27714

Sturm suite [20]:
Degree 21:
Time: 0.016980 s

Coeff numerator bits: 27831

Degree 20:

Time: 0.018966 s

Coeff numerator bits: 31368

Sturm suite [21]:

Degree 20:

Time: 0.018790 s

Coeff numerator bits: 31499

Degree 19:

Time: 0.022757 s

Coeff numerator bits: 35279

Sturm suite [22]:

Degree 19:

Time: 0.021711 s

Coeff numerator bits: 35396

Degree 18:

Time: 0.024689 s

Coeff numerator bits: 39420

Sturm suite [23]:

Degree 18:

Time: 0.023606 s

Coeff numerator bits: 39538

Degree 17:

Time: 0.027107 s

Coeff numerator bits: 43799

Sturm suite [24]:

Degree 17:

Time: 0.026480 s

Coeff numerator bits: 43926

Degree 16:

Time: 0.028661 s

Coeff numerator bits: 48408

Sturm suite [25]:

Degree 16:

Time: 0.027305 s
Coeff numerator bits: 48512

Degree 15:
Time: 0.030690 s
Coeff numerator bits: 53239

Sturm suite [26]:
Degree 15:
Time: 0.030303 s
Coeff numerator bits: 53362

Degree 14:
Time: 0.033536 s
Coeff numerator bits: 58285

Sturm suite [27]:
Degree 14:
Time: 0.032802 s
Coeff numerator bits: 58372

Degree 13:
Time: 0.037692 s
Coeff numerator bits: 63509

Sturm suite [28]:
Degree 13:
Time: 0.034659 s
Coeff numerator bits: 63601

Degree 12:
Time: 0.036999 s
Coeff numerator bits: 68914

Sturm suite [29]:
Degree 12:
Time: 0.037741 s
Coeff numerator bits: 68985

Degree 11:
Time: 0.038775 s
Coeff numerator bits: 74439

Sturm suite [30]:

Degree 11:
Time: 0.036931 s
Coeff numerator bits: 74522

Degree 10:
Time: 0.039907 s
Coeff numerator bits: 80113

Sturm suite [31]:
Degree 10:
Time: 0.040671 s
Coeff numerator bits: 80163

Degree 9:
Time: 0.040016 s
Coeff numerator bits: 85881

Sturm suite [32]:
Degree 9:
Time: 0.038419 s
Coeff numerator bits: 85930

Degree 8:
Time: 0.038916 s
Coeff numerator bits: 91721

Sturm suite [33]:
Degree 8:
Time: 0.036237 s
Coeff numerator bits: 91741

Degree 7:
Time: 0.039640 s
Coeff numerator bits: 97599

Sturm suite [34]:
Degree 7:
Time: 0.035829 s
Coeff numerator bits: 97531

Degree 6:
Time: 0.038414 s
Coeff numerator bits: 103342

Sturm suite [35]:
Degree 6:
Time: 0.033164 s
Coeff numerator bits: 103298

Degree 5:
Time: 0.034571 s
Coeff numerator bits: 108999

Sturm suite [36]:
Degree 5:
Time: 0.030651 s
Coeff numerator bits: 108820

Degree 4:
Time: 0.032505 s
Coeff numerator bits: 114303

Sturm suite [37]:
Degree 4:
Time: 0.026877 s
Coeff numerator bits: 114067

Degree 3:
Time: 0.028983 s
Coeff numerator bits: 119137

Sturm suite [38]:
Degree 3:
Time: 0.022048 s
Coeff numerator bits: 118634

Degree 2:
Time: 0.023398 s
Coeff numerator bits: 122928

Sturm suite [39]:
Degree 2:
Time: 0.016907 s
Coeff numerator bits: 122211

Degree 1:
Time: 0.018079 s
Coeff numerator bits: 125248

Degree:40 [-50,50]

```
Degree 1:  
Time: 0.008483 s  
Coeff numerator bits: 1  
  
Time cost for function bound: 216.683123 s  
result correct
```

Number of divided interval

	40,-50,50	(40,-50,50)	(40,-100,100)	(30,-100,100)
2	296.63196	299.683366	507.807592	55.735257
3	215.099607	212.924295	332.885513	40.568368
4	341.635973	337.60473	577.212364	65.925093
5	245.97292	237.12674	406.905787	48.37168
6	414.818114	402.245508	677.704455	83.445564
7	298.403678	284.561404	483.780387	54.239788
8		473.427537	835.282763	89.814903
9		316.727339	541.129935	58.10542
10		546.544642		102.953775
nbRacine	879.661508	778.23831		131.175104